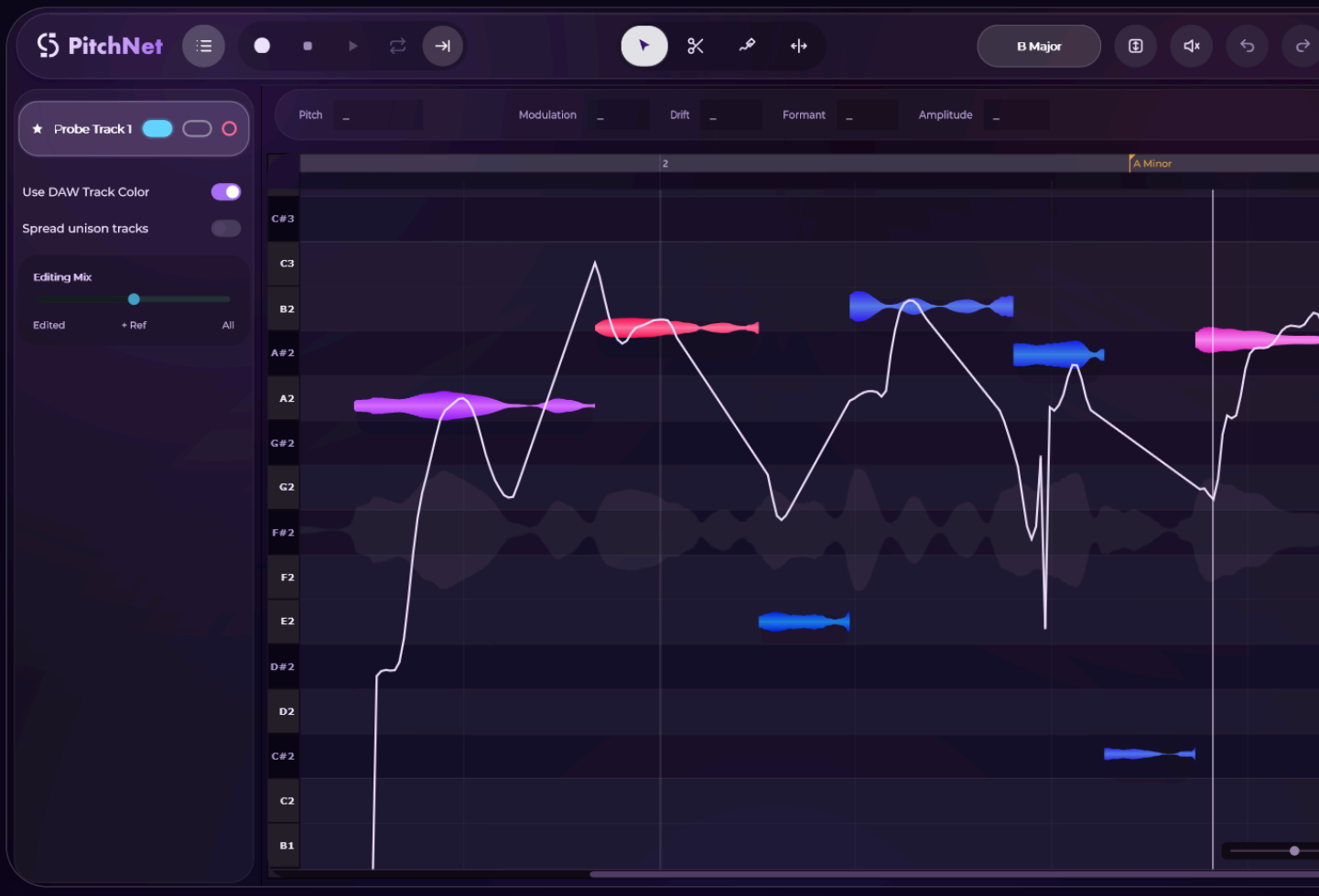
PitchNet

Stars Clique Edition

User Manual

Every tool, panel and trick of the neural vocal pitch editor: recording from FL Studio, editing several tracks at once, tuning backing vocals, keys and custom scales.



Version 1.0 · VST3 for Windows

PitchNet by Session Loops and contributors, based on HachiTune

Stars Clique edition: modified by Goth Angel (Mikhail Mikhalchenko)

github.com/9519860664r-oss/pitchnet-stars-clique

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CHAPTER 1

What is PitchNet Stars Clique Edition

PitchNet is a pitch editor for vocals. It finds the notes in a recording, shows them as blobs on a piano roll and lets you move, stretch, tune and reshape them. A neural vocoder renders the result in real time, so edits sound natural.

PitchNet was created by **Session Loops and contributors** and is based on **HachiTune** by the HachiTune contributors. The pitch detection, note segmentation, neural vocoder and the editor itself are their work.

The **Stars Clique Edition** is a modified version by Goth Angel (Mikhail Mikhalchenko), Stars Clique. It adds working with many vocal tracks in hosts without ARA, such as FL Studio, tools for backing vocals and tuning, and a new look.

Multitrack in any host

Every PitchNet on every mixer channel appears in one track list. Edit as many tracks at once as you like, show the rest as grey ghosts.

Backing vocals to the lead

Align timing to a reference track, tune to it in unison or in an interval, level loudness, quantize time.

Recording from the DAW

Transfer records from the DAW, with punch-in, re-recording any section and replace ranges.

Keys and tuning

A key shared by all tracks, key changes along the song, A4 from 400 to 480 Hz and your own scales with cents.

Neural engine

- **FCPE** detects pitch, **GAME** splits the audio into notes.
- **PC-NSF-HiFiGAN** resynthesizes the edited voice. Unpitched sounds (breaths, consonants, rasp) stay as in the original.
- All models run locally on your computer. Nothing is sent anywhere.

CHAPTER 2

Installation

Requirements

- Windows 10 or 11, 64-bit.
- A VST3 host: FL Studio, REAPER, Cubase, Studio One, Ableton Live, Bitwig and others.
- About 400 MB of disk space (the models take most of it).

Install

1. Download **PitchNet-StarsClique-1.0.0-Setup.exe** from the Releases page on GitHub.
2. Close your DAW, then run the installer and accept the license (AGPLv3).
3. Choose the components: the **VST3 plug-in** and, if you want it, the **standalone app**. The neural models are always installed.
4. Open your DAW and rescan plug-ins. Look for **PitchNet (Stars Clique ver)**; the host lists Session Loops as its maker.

What	Where it goes
VST3 plug-in	C:\Program Files\Common Files\VST3\Stars Clique\
Standalone app	C:\Program Files\Stars Clique\PitchNet\
Neural models	C:\ProgramData\Session Loops\PitchNet\models\
Your settings, skin, saved scales	%APPDATA%\PitchNet\

> TIP
> The models folder keeps the original PitchNet name on purpose: the original PitchNet and this edition share one copy of the models.

Update and remove

To update, run the newer installer over the old one. To remove, use **Settings › Apps › PitchNet Stars Clique Edition › Uninstall**. Your settings in %APPDATA% stay, so a later install picks them up.

CHAPTER 3

First steps in FL Studio and REAPER

PitchNet needs to hear the audio before it can edit it. In hosts with ARA it reads the audio directly; elsewhere it records it while the DAW plays. This is called **Transfer**.

FL Studio (and other hosts without ARA)

1. Put **PitchNet (Stars Clique ver)** on the mixer insert of each vocal track you want to edit. One instance per track.
2. Open any of the PitchNet windows. The track list on the left shows every instance with its mixer channel name.
3. Click the pink **Transfer** ring ◯ on the tracks you want to record (or the record dot in the toolbar for the track of this window).
4. Play the song in FL. PitchNet records the armed tracks, all in one pass.
5. Stop or pause. Each take is finished and analysed; its notes appear in a few seconds.

> TIP
> If FL loops a selection, the jump back to the start of the loop also finishes the take. Playing again from any point re-records from there and replaces the audio of that section only.

REAPER (ARA)

1. Add PitchNet as an FX on a media item or track. REAPER opens it in ARA mode.
2. The audio is available at once: no recording needed. The notes appear after analysis.
3. Edits play back through REAPER as usual.

Hearing the result

PitchNet replaces the original audio with the edited one in real time, so you hear the edits through your DAW's playback. In the plug-in window, **Live audition** (the speaker button) lets you hear a note while you drag it.

CHAPTER 4

The window at a glance



Toolbar on top, track list on the left, Note Inspector above the editor.

Toolbar, from left to right

Control	What it does
PitchNet logo	Menu: change skin, export MIDI, user manual and releases, credits.
Track list	Shows or hides the track list.
Record (pink dot)	Transfer for this window's track (hosts without ARA). Glows while armed.
Stop, Play / Pause, Loop	Transport. In hosts without ARA the DAW drives playback, so these may be disabled.
Follow	The view follows the playhead. It pauses while notes are selected or after you scroll by hand.
Tools	Main tool, Split, Pitch drawing, Timing (keys 1 – 4).
Key (for example "B Major")	Key, mode, custom scales and the A4 reference pitch. Shared by all tracks.
Correct Pitch	Opens the Correct Pitch macro for the selected notes.

Live audition	Hear notes while you drag them.
Undo, Redo	Ctrl+Z, Ctrl+Y.
Clip / Track (ARA only)	Edit one clip or the whole track.
Parameters	Shows or hides the parameters panel on the right.

Editor

- **Piano keys** on the left, the time ruler and the loop strip on top.
- **Notes** are drawn as blobs shaped like their waveform. The white line is the pitch curve.
- **Note colour shows how well a note hits the key**: blue exactly on a note of the key, then violet, pink and red the further it is from it (red at half a semitone or more). A note on a semitone outside the key is red. With a custom scale the cents count too.
- **Rows follow the key**: notes of the key are light rows, the rest dark, like a scale highlight in a DAW piano roll. Change the key and the rows change with it; key changes along the song are shown too. With a custom scale, a dashed line marks each detuned note's exact pitch.
- Right-click the background to switch to **Pitch Lines**.
- Zoom sliders and the overview button sit in the bottom-right corner.

CHAPTER 5

Tracks: Edit, Ghost, Transfer

Each PitchNet instance is a track. The track list shows all of them, in every window, with the name and colour from the DAW.

Control	Meaning
▭ window mark	The track this window belongs to.
★ star	The lead of a group of doubles, in the group's colour.
D	A double, in its group's colour; a stripe of that colour runs down the row.
Highlighted row	A selected track. Click a row to select it, Ctrl +click to add or remove one, Shift +click to select a range.
Edit (blue pill)	The track's notes are shown in colour and can be edited. Any number of tracks can be on at once.
Ghost (white pill)	The track's notes are shown as grey blobs for orientation only. The closer a ghost note is to the key, the lighter it is.
◯ Transfer (pink ring)	Arms the track for recording (hosts without ARA). Filled while armed, with a halo while audio is being recorded.

Editing several tracks

- Turn on Edit for several tracks: their notes are edited together (selection, pitch, the arrow keys, timing, delete, macros).
- In REAPER, Edit on another track shows its clips on the timeline next to the active track's (the view switches to Track).
- Select several tracks, then press Edit, Ghost or Transfer on one of them: every selected track changes the same way. A button on a track that is not selected changes only that track.
- Your own track (window mark), and in REAPER the track the window shows, can be switched off or made a ghost like any other.
- Double-click a ghost note to swap the roles of the two tracks.
- When you click a note, the other tracks turn grey for a moment, so you see which track it belongs to.
- Changes made in one window appear in the others right away.

Lead and doubles: unison

Doubles usually repeat the lead in places. PitchNet can tune them to the lead note for note.

1. **Make a group.** Select the lead and its doubles (click, Ctrl +click) and press **Group selected**. This window's track leads if it is selected, otherwise the top one; click a double's **D** to make it the lead instead. Or press **Find doubles**: PitchNet compares the tracks' notes and suggests groups; nothing changes until you press Create.
2. **Press Unison** under the list. It lights up while on: the doubles take the lead's notes. Each double keeps what makes it a separate singer (its own vibrato, jitter, slides and transitions between notes) and takes from the lead only where the notes sit and how they drift. Lead and double are compared at the moments they sing the same sound, so a double sung a little early or late still gets the right note, and the correction changes smoothly at the double's own transitions, without clicks. Octave doubles stay in their octave; the lead's sibilants never set a double's pitch. Where the lead is silent (ad-libs), or a double note is more than about 3.5 semitones from the lead's (another syllable, not a missed note), the double is left alone.
3. **Set the strength** with the slider: 0 % leaves the doubles as sung, 100 % on the lead's notes. Even at 100 % the last few cents stay the double's own, so stacked doubles sound like people, not copies.

- While Unison is on, the doubles follow every change to the lead, even with their windows closed.
- A double note follows the lead note it sounds with the longest, when they sound together for at least a quarter of the double note. Splits and joins line up: inside the note the double follows the lead's pitch frame by frame.
- Switch Unison off, or take a track out of the group, and the doubles go back to their own pitch.
- **Ungroup** (or right-click › **Remove from group**) takes every selected track out of its group; taking out a lead ends its group. Right-click also offers **Add as double to**. Groups are saved with the DAW project.

Lead and doubles: timing

Press **Timing** in the group, under Unison, and the doubles line up with the lead in time.

- The doubles are matched to the lead by the shape of the sound (its spectrum, loudness and onsets), syllable by syllable, not only by where notes start. Inside a note the sound stretches or squeezes where a syllable is longer or shorter than the lead's.
- The slider sets the strength: 0 % leaves the timing as sung, 100 % follows the lead. Differences under about 23 ms always stay, so the doubles still sound sung, and no sound moves more than about 300 ms.
- Sibilants move along without stretching. Phrases are aligned one by one; long pauses are left out.
- Timing is lined up first, then Unison tunes the pitch, so both follow the same syllables. Both work live, even with the doubles' windows closed.
- Switch Timing off and every note goes back to its own timing.
- In REAPER groups, Unison and Timing work the same way, and a group is saved with the REAPER project. REAPER lets PitchNet read the other tracks' audio only after the song has played once: if the window says *Press Play once*, start playback for a moment.

Settings under the list

- **Use DAW Track Color**: rows and notes use the colour of the mixer channel.
- **Spread unison tracks**: notes of different tracks that sit on top of each other are fanned out, so you can grab each one.
- **Editing Mix**: when PitchNet plays a preview with the DAW stopped: left = only edited tracks, centre = with the ghosts, right = every track.

CHAPTER 6

Recording with Transfer

In hosts without ARA, PitchNet records the audio as the DAW plays it. You can record many tracks in one pass and fix single sections later.

Takes

- **Arm** one or more tracks with the pink ring, then play.
- **Pause or stop** in the DAW: every armed track finishes its take and is analysed.
- **Loop end**: when the DAW jumps back to the start of a looped selection, the take finishes too.
- **Punch in / out**: switch Transfer on or off during playback to record only part of the song.
- **Re-record** from any point: the new audio replaces only the section you recorded again; your edits elsewhere stay.

Replace ranges

Right-click the time ruler for the replace-range menu.

- **Show Replace Ranges**: shows on the ruler which sections PitchNet plays instead of the original.
- Drag the edges of a range to make it longer or shorter.
- **Set Replace Ranges to Notes**: PitchNet only replaces the audio where there are notes.
- **Reset Replace Ranges to Transferred Audio**: back to the whole recorded section.

> NOTE
> Transfer records what reaches the plug-in. Put PitchNet before effects such as reverb or delay, so it gets the dry vocal.

CHAPTER 7

Editing notes

Main tool 1

- **Drag a note** up or down to change pitch, sideways to move it in time. The first movement locks the axis, so a pitch change does not shift the timing by accident.
- **Hold Alt** while dragging to move without the grid.
- **Double-click** a note: it snaps to the nearest semitone, or to the nearest note of the key when the pitch snap is set to the scale (custom scales with their cents). With Alt it keeps its offset.
- Drag the **edges** of a note to make it longer or shorter.
- Moving to another beat keeps the note's offset from the beat.

Split 2, Pitch drawing 3, Timing 4

- **Split**: click inside a note to cut it in two.
- **Pitch drawing**: draw the pitch curve by hand, for slides and vibrato.
- **Timing**: move and stretch notes in time.
- Note boundaries can be soft (linked edges with a smooth transition) or hard.

Sibilants

- PitchNet finds the sibilants of every note: "s", "sh", "f", "t", "k" and breaths, the parts with no pitch. They sit at the start and end of a note and are drawn **hatched**; a breath can be a hatched note of its own.
- A note moves with its sibilants, but they keep their own sound: when the pitch changes, only the pitched part is shifted; when the note is moved or stretched, the sibilants move without stretching ("sweet" becomes s-weeee-t).
- Quantize Time and Align to Reference place a note's musical start after its sibilant.
- Right-click the editor › **Sibilants...**: **Show Sibilants**, **Find Sibilants Again**, **Sibilant from Note Start to Here** and **from Here to Note End** (at the point you right-clicked), **Remove Sibilants** (they are then pitch-shifted with the note, as an effect), and **Sibilant Balance** from -100 % (sibilants muted) to +100 % (only the sibilants). Softer sibilants on doubles keep a group from hissing in chorus. The commands work on the selected notes, or on the note you clicked.

Note Inspector

The strip above the editor shows the selected notes. Type a value and press Enter to set it for every selected note. If the notes differ, the field shows "–".

Field	Input

Pitch	A note name (`C3`, `F#4 +10`) or a change (`+2`, `-30 ct`). The frequency in Hz is shown next to it.
Modulation	Vibrato depth in %. 0 % flattens it, 100 % is the original.
Drift	Slow pitch drift in %.
Formant	Formant shift in semitones: the colour of the voice.
Amplitude	Note level in dB.
Note Off (round button)	Mutes the selected notes; click again to bring them back. A muted note shows only its outline.

Selecting notes

- `Ctrl`+click adds or removes a note, `Shift`+click selects a range, `Ctrl+Shift`+drag draws a selection box.
- Click a key on the piano to select all notes of that pitch; double-click it for every octave.
- Right-click menu: **Select All**, **Select Same Pitch**, **Select All Following**, **Invert Selection**, **Restore Last Selection**.

Copy, cut, paste

- `Ctrl+C`, `Ctrl+X`, `Ctrl+V` copy notes with their audio.
- Paste goes to the cursor; over a selection it replaces the selected notes and stretches to fit.
- Copy between tracks: one to one, many to one, or one to many.

Undo and restore

- `Ctrl+Z` / `Ctrl+Y` undo and redo any edit.
- Right-click › **Restore Original**: pitch, timing, amplitude, formant or all edits of the selected notes.
- **Whole Track: Undo All Changes** resets the whole track.
- **Add Random Deviations**: a little natural looseness in pitch or time (subtle, medium, strong).
- `Delete` or `Backspace` mutes the selected notes.

CHAPTER 8

Macros

Macros work on the selected notes (or all notes when nothing is selected). They remember their settings and leave notes you edited by hand alone if you ask them to.

Correct Pitch

Pulls the pitch centre of notes towards their target and evens out drift, with separate strengths. With **Snap to Scale** on, the target is the nearest note of the key in force at the note (key changes, custom scales with cents); off, it is the nearest semitone. Both follow the A4 reference pitch. Option: include notes edited by hand. Opened from the toolbar button.

Quantize Time

Moves notes to the grid. Groove reference: Auto, a note value from 1/1 to 1/32 with triplets, or another track. Intensity sets how far notes move.

Align to Reference

Lines backing vocals up with a lead track: the starts and ends of notes, and optionally pitch and loudness, with a strength control. It can also tune to the reference in unison or in an interval of the key (third, fourth, fifth, octave up or down).

Leveling

Two sliders bring loud and quiet notes towards the average level.

Quantize Time, Align to Reference, Leveling and Add Random Deviations are in the right-click menu of the editor.

CHAPTER 9

Key, tuning and custom scales

One key for every track

The key in the toolbar is shared by every track and every window. Change it once and the grid, the ghost colours and Correct Pitch follow everywhere.

Key changes

- Right-click the time ruler › **Key change here**. From that point on, the new key applies (grid, ghost notes, Correct Pitch).
- Right-click a marker to remove it, or remove all key changes at once.

Reference pitch (A4)

Key menu › **Reference pitch (A4)**: any value from 400 to 480 Hz. Shared by every track.

Custom scales

1. Key menu › **Edit custom scale…**
2. Choose which notes belong to the scale and detune any of them in cents.
3. Give it a name and **Save**. Saved scales appear under **Saved custom scales** in the key menu.
4. Press **Use**: double-click snapping and Correct Pitch now tune to your scale and its cents.

> TIP
> Saved scales live in %APPDATA%\PitchNet\custom_scales.json. Copy the file to share your scales.

CHAPTER 10

Navigation and zoom

Move around the editor with the mouse and a few modifier keys.

Action	Result
Mouse wheel	Scroll up and down
Shift + wheel	Scroll left and right
Ctrl + Alt + wheel, or pinch	Zoom both axes around the mouse
Ctrl + Shift + drag	Hand: move the view
Ctrl + Alt + drag	Zoom tool around the point you clicked
Drag up/down on the time ruler	Zoom in time; drag sideways to move the cursor
Double-click a scroll bar	Show everything (or the loop range)
Ctrl + Shift + double-click a note	Zoom to that note or the selection
Ctrl + Alt + double-click the background	Back to the previous zoom
Zoom sliders (bottom right)	Horizontal and vertical zoom

CHAPTER 11

Keyboard shortcuts

Keys	Command
1 2 3 4	Main tool, Split, Pitch drawing, Timing
Space	Play / pause
Esc	Stop
Home / End	Go to start / end
Ctrl+Z	Undo
Ctrl+Y, Ctrl+Shift+Z	Redo
Ctrl+A	Select all notes
Ctrl+C / Ctrl+X / Ctrl+V	Copy / cut / paste notes
Up / Down	Selected notes a semitone up / down
Shift + Up / Down	An octave up / down
Delete, Backspace	Mute the selected notes
Ctrl+Shift+D	Show or hide the delta pitch curve
Ctrl+Shift+B	Show or hide the base pitch curve
Ctrl+,	Settings (standalone app)

CHAPTER 12

Skins, export and settings

Skins

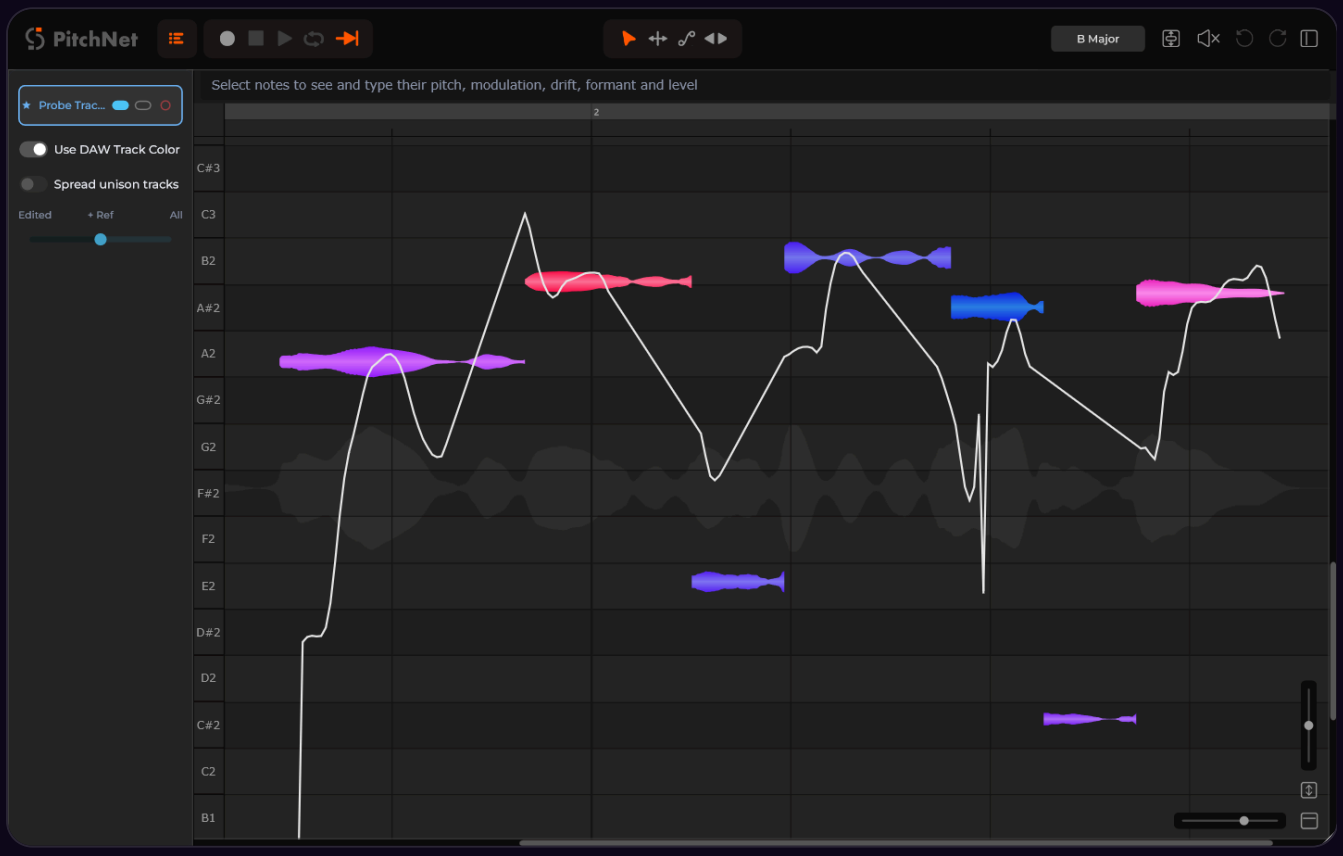
Click the PitchNet logo › **Change skin**. The last skin you chose opens next time.

Stars Clique

Deep violet with floating glass panels, vector icons and soft glows.

Classic

The original dark grey PitchNet look.



The Classic skin.

Export

- Logo menu › **Export MIDI...**: the notes of the shown tracks, each track on its own MIDI channel.
- The edited audio itself comes out of your DAW: render or bounce the track as usual.

Parameters panel

The button on the far right of the toolbar opens the parameters panel: pitch and timing settings, rendering options, language and interface brightness.

CHAPTER 13

Troubleshooting

The plug-in does not appear in the DAW

Rescan VST3 plug-ins. In FL Studio: Options › Manage plugins › Find more plugins. Check that the plug-in is in C:\Program Files\Common Files\VST3\Stars Clique\.

No notes appear after recording

Make sure the track was armed (pink ring filled) while the DAW played, and that playback was stopped or paused afterwards. Analysis takes a moment after the take ends.

PitchNet says the models are missing

Run the installer again: the models must be in C:\ProgramData\Session Loops\PitchNet\models\.

I hear the original instead of my edits

The edited audio replaces the original only inside the replace ranges. Right-click the time ruler › Show Replace Ranges.

Reverb or delay is recorded too

Move PitchNet before time-based effects on the mixer channel.

Reporting a problem

Open an issue on GitHub with your host, its version and the steps that lead to the problem.

CHAPTER 14

Credits and license

PitchNet · original authors

Created by **Session Loops and contributors**: Jie Chen, 一支粉笔, KRKT, hrukalive, Steven Cooper, hyperrealist, 暗雨冥.
github.com/SessionLoops/PitchNet

HachiTune

PitchNet is based on HachiTune by the HachiTune contributors.
github.com/KCKT0112/HachiTune

Stars Clique Edition · modifications

Modified and extended by **Goth Angel (Mikhail Mikhalchenko)**, Stars Clique.
github.com/9519860664r-oss/pitchnet-stars-clique

Models and libraries

PC-NSF-HiFiGAN and GAME by Team OpenVPI · FCPE by CN_ChiTu · RMVPE by yxlllc · JUCE · ONNX Runtime · r8brain-free-src by Aleksey Vaneev · ARA SDK by Celemony Software GmbH and contributors.

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